Physical Chemistry Cume

$C_{p,m}$ (H_2O, liquid) = 75.3 $JK^{-1}mol^{-1}$; $C_{p,m}$ (H_2O, gas) = 33.6 $JK^{-1}mol^{-1}$; both can be considered independent of temperature. $\Delta_{vap}H$ (H_2O, 373K) = 44.0 $kJmol^{-1}$

1. (50 pts)

The excess energies (relative to the bulk) of the various nanoparticles at 298K are calculated by dissolving the hydrated nanoparticles in a molten oxide melt ($3Na_2O_4$-MoO_3) at 973K.

e.g.

$Co_3O_4.nH_2O$ (nano, 298K)→Co_3O_4 (dissolved in $3Na_2O_4$-MoO_3, 973K) + nH_2O (gas, 973K)

a) How do they calculate the surface energy from the excess enthalpy?
b) How are they able to calculate the excess enthalpy of the anhydrous nanoparticles at 298K?
c) The water adsorption energy for Co_3O_4 at 298K was measured to be -93.1 $kJmol^{-1}$. For a given particle size, derive an expression for the difference in excess enthalpy between anhydrous Co_3O_4 and hydrous Co_3O_4 which depends only on the number of moles of water, n, given off in the above reaction?
d) Why does this difference depend on nanoparticles size?

2. (30 pts)

Table 1. Surface energies for transition metal oxides and related systems.

Phase	Surface energy (anhydrous surface) (J/m^2)	Surface energy (hydrated surface) (J/m^2)
Co (hcp)	2.22 ± 0.30 (*9*)	1.66 ± 0.23*
Fe (bcc)	2.12 ± 0.29 (*9*)	1.59 ± 0.22*
$Fe_{0.947}O$ (rocksalt)	[3.6]†	[2.8]†
CoO (rocksalt)	3.57 ± 0.30	2.82 ± 0.20 (*6*)
NiO‡ (rocksalt)	3.5 ± 0.5	2.4 ± 0.4
Co_3O_4 (spinel)	1.96 ± 0.05	0.92 ± 0.04
Fe_3O_4‡ (spinel)	1.44 ± 0.30	0.79 ± 0.28
Mn_3O_4‡ (spinel)	—	0.5 ± 0.4
γ-Fe_2O_3 (spinel)	0.71 ± 0.13 (*10*)	0.57 ± 0.10 (*10*)
γ-Al_2O_3 (spinel)	1.53 ± 0.40 (*25*)	0.7 ± 0.4 (*25*)
$MgAl_2O_4$ (spinel)	1.8 ± 0.3§	0.9 ± 0.3§
Fe_2O_3 (hematite)	1.9 ± 0.3 (*26*)	0.75 ± 0.16 (*27*)
Mn_2O_3‡ (bixybite)	—	1.66 ± 0.21

*Energy of hydrated metal surface is assumed to be 0.75 that of the anhydrous surface, as discussed in the text. †The similarity of values for CoO and NiO suggests that it is reasonable to assign a similar surface energy to wustite, $Fe_{0.947}O$. Changes in the oxygen nonstoichiometry of wustite at the nanoscale are not considered. Neglecting such variations does not significantly change the positions of the redox equilibria. ‡For calorimetric data, see table S1. §Calculated using data from previous study (*11*).

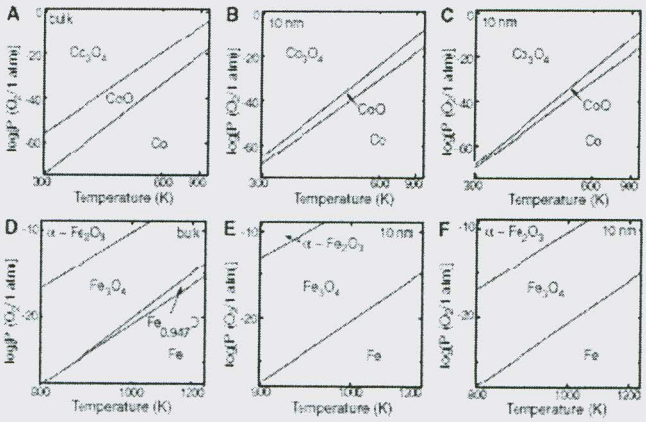

Fig. 2. Calculated phase diagrams for bulk and 10-nm-diameter spherical particles (**A**) Co-O (bulk). (**B**) Co-O (anhydrous 10-nm particles). (**C**) Co-O (hydrated 10-nm particles). (**D**) Fe-O (bulk). (**E**) Fe-O (anhydrous 10-nm particles). (**F**) Fe-O (hydrated 10-nm particles). The temperature scale is in 1/T, in order to make a linear plot, although the labeled temperatures are in T (kelvin). There is no stability field for 10-nm wustite $Fe_{0.947}O$.

Explain the thermodynamic basis for the following observations in the phase diagrams

a) The stable range of CoO becomes much narrower as we move from bulk to 10 nm particle
b) $Fe_{0.947}O$ Is not observed in 10 nm particles
c) Hydrated α-Fe_2O_3 is stable over a wider range than anhydrous Fe_2O_3

d) The authors also observed that "CoO nanoparticles smaller than 8 nm when dropped into water, evolved a gas that was shown to be hydrogen by its flammability ... and was not observed for larger particles." (i) Why was this reaction not observed for larger particles? (ii) Write a balanced chemical reaction for what you think is occurring in the 8 nm particles?

3. (20 pts)

The authors discussed the fact that an earlier paper had reported that Co_3O_4 makes an excellent catalyst for low temperature CO oxidation. How can the results from this paper be used to design catalysts for this purpose?